

Network Intrusion Detection Using Supervised Machine Learning For Ddos Attack Classification

¹Neha Shireen, ²Hafsa Tasneem

^{1,2}Assistant Professor, ^{1,2}Department of CSE-AIML,

^{1,2}Lords Institute of Engineering and Technology, Hyderabad, India

Abstract— The rapid growth of internet-based services has significantly increased the exposure of network infrastructures to sophisticated cyber threats, particularly Distributed Denial-of-Service (DDoS) attacks that disrupt service availability and compromise organizational operations. Traditional signature-based intrusion detection systems (IDS) struggle to identify evolving and zero-day attacks, necessitating intelligent and adaptive detection mechanisms.

This paper proposes a supervised machine learning-based intrusion detection framework for accurate classification of malicious network traffic. Traffic data from the KDD Cup 99 benchmark dataset are preprocessed using feature selection and normalization techniques to improve model efficiency and reduce computational overhead. Multiple classifiers, including C4.5 Decision Tree and Naïve Bayes, are trained and evaluated for attack detection performance. Experimental results demonstrate that the Decision Tree model achieves superior detection accuracy and lower false-positive rates compared to probabilistic approaches.

Index Terms—Distributed denial of service, Machine learning, Random Forest, Decision Tree, Intrusion Detection System.

I. INTRODUCTION (HEADING 1)

In With the rapid expansion of information technology, computer networks are now widely used across businesses, industries, and everyday life. This makes it crucial for IT administrators to ensure network security and trustworthiness. However, the rapid growth of technology has also led to challenges in establishing reliable networks, which is a complex task. Various types of cyberattacks threaten the privacy, integrity, and availability of computer networks, with denial-of-service (DOS) attacks being particularly common. These attacks aim to disrupt multiple end-user services by overwhelming network resources and overloading systems. In recent years, there has been growing interest in using machine learning techniques, particularly classification algorithms, to build more accurate intrusion detection models. This paper proposes leveraging machine learning classifiers such as Naive Bayes and Decision Tree classifiers to enhance the accuracy of intrusion detection systems using a Knowledge Discovery in Databases (KDD) dataset. This paper conducts experiments to evaluate different machine learning classifiers using the KDD intrusion dataset, assessing their performance based on various metrics. Specifically, it focuses on improving the detection rate of the intrusion detection system by analysing false positive and false negative metrics. [1] The KDDCup 1999 intrusion detection dataset is crucial for refining intrusion detection systems and is extensively utilized by researchers in the field. In addition to analysing the KDDCup'99 dataset as well as providing an extensive review of the classification methods used in intrusion detection, this paper gives an overview of intrusion detection. [2] By relaxing the conditional independence requirement, the Hidden Naive Bayes (HNB) model differs from the conventional naive Bayes method. This paper focuses primarily on the Hidden Naive Bayes model, and preliminary findings suggest that it outperforms traditional naive Bayes models in terms of error rate, misclassification cost, and accuracy, particularly when applied to the Knowledge Discovery and Data Mining (KDD) Cup 1999 dataset. [3] To address the multitude of attacks targeting network resources, intrusion detection systems (IDS) have become essential. IDS monitors network traffic to detect and alert administrators to any suspicious activity or potential attacks. They can focus on monitoring traffic for specific devices (host-based IDS) or all network traffic (networkbased IDS), with the latter being more common. There are two main types of IDS: anomaly-based and exploit-based. Anomaly-based IDS detect deviations from normal behaviour by comparing current traffic patterns with historical data, while exploit-based IDS rely on predefined attack signatures. Both types have their advantages and drawbacks, leading to challenges in accurately detecting all types of attacks. This paper presents a newly collected dataset that includes modern DDoS attacks across various network layers, including SIDDoS and HTTP Flood, which were not commonly available in existing datasets. The study

evaluates the performance of three popular classification techniques: Naïve Bayes, Random Forest, and Multilayer Perceptron (MLP). [4] This paper proposes a hierarchical, layered approach to enhance the detection rate of both majority and minority attacks. For each attack class, the proposed approach uses a Naive Bayes classifier with a K2 learning procedure on a NSL KDD dataset. With this approach, every layer is shown separately to identify a certain kind of attack category. The layer receives the output from the previous layer, which helps to increase the detection rate and better classify majority and minority attacks. [5] Given the complexity of distinguishing normal and attack packets, machine learning algorithms are increasingly being explored as an alternative approach for intrusion detection. These algorithms automate the process of identifying patterns of normal and intrusive behaviour, reducing the need for manual intervention. This paper introduces a novel learning algorithm for adaptive network intrusion detection, employing a naive decision tree and Bayesian classifier. It aims to achieve balanced detections while maintaining acceptable false-positive rates across various network attack types. Additionally, the algorithm tackles challenges in data mining, including reducing noise in training data, managing missing attribute values, and handling continuous attributes. [6] This paper introduces a traffic classification scheme aimed at enhancing classification accuracy in scenarios with limited training data. It proposes a method to combine the predictions of naive Bayes for different traffic flows. [7]

Literature Review

In this paper, we explore the effectiveness of various machine learning classifiers for intrusion detection using the KDD intrusion dataset. Our experiments aimed to evaluate performance metrics like false negatives and false positives to improve detection rates. We found that the decision table classifier had the lowest false negative rate, while the random forest classifier achieved the highest average accuracy. This research underscores the importance of robust intrusion detection systems against evolving cyber threats. [1] As security becomes increasingly crucial for organisational information systems, detecting threats and attacks on computer networks has become a pressing issue. Intrusion detection systems (IDS) have emerged as essential tools to safeguard computer systems from various attacks. In recent years, data mining-based intrusion detection has gained significant attention from researchers, who propose different algorithms across various categories. Constructing effective classifiers presents another challenge in building Network-mimicking DDoS attacks can also be stopped with detecting methods based on chaos theory. The work of Ashley Chonka et al. [14] describes a way to find network-imitating DDoS attacks that is based on chaos theory. The suggested method looks at how network traffic behaves randomly to accurately tell the difference between valid and attack traffic. It can be hard to keep an eye on application-layer DDoS attacks on famous websites because web applications and protocols are so complicated. In their paper [15], Yi Xie et al. describe a way to keep an eye on application-layer DDoS strikes that are aimed at popular websites. For finding and stopping application-layer threats, the study stresses how important it is to use real-time tracking and anomaly detection methods [16].

Using network self-similarity to find DDoS attacks is another way to protect yourself before they happen. Network self-similarity research [17] suggest as a way to find DDoS attacks. The method measures the self-similarity of network data to look for changes from normal behavior that could be signs of DDoS attacks. It is very important to be able to quickly find and stop DDoS and TCP flood attacks in cloud settings. Experts in [18] show a good way to find and stop DDoS TCP flood attacks in cloud settings. The system uses network traffic analysis and machine learning to find and stop TCP flood attacks in real time. To sum up, the literature review shows the wide range of approaches and tools that are used to find and stop DDoS attacks. Researchers keep coming up with new ways to make networks safer and more resistant to new cyber threats [19]. These include flow based analysis, entropy-based measures, and machine learning algorithms.

II. METHODOLOGY

a) *Proposed Work*

We want to use the Random Forest method in our suggested work as a powerful way to find and stop Distributed Denial of Service (DDoS) attacks. We expect to be able to tell the difference between normal network traffic and harmful intrusions very accurately by using its ability to handle big datasets and find complex patterns. On both the academic and practical levels, this study is likely to make a big difference. In theory, we want to give new information about how Random Forest can be used to find DDoS attacks. This could lead to new discoveries in feature extraction and model training. In real life, our goal is to create a real and effective way to protect against DDoS attacks, which will make the internet safer and more robust. Because DDoS attacks are so disrupting and real-time discovery and prevention are hard, our suggested solution has a lot of potential to protect network security and lower the risks that cyber threats could bring.

b) **System Architecture:**

DDoS Attack Detection Using Random Forest in Machine Learning starts with gathering a variety of datasets that include network traffic data. This makes sure that both regular and attack situations are represented. Once the data has been cleaned up and any missing or unusual values have been taken care of, it is labeled as either normal or a DDoS attack based on trends that have already been found. Information gain and other feature selection methods help find the best set of useful features. Using the preprocessed and feature-selected dataset to train a Random Forest model is what the system is all about.

Ensemble learning is used to improve accuracy and stability. To get the best results from hyperparameters, they are carefully tested using validation datasets and measures such as accuracy and recall. The Random Forest model is put into production after it has been trained, and it works perfectly with the current network security infrastructure. Continuous tracking makes sure that it keeps working, and changes are made on a regular basis to keep up with changing DDoS attack trends. The architecture map shows all the parts of the system clearly, making sure that a complete plan is made to improve network security. efficient intrusion detection systems. The KDDCup 1999 intrusion detection dataset serves as a cornerstone for refining such systems and is widely utilized by researchers in this field. This paper provides an overview of intrusion detection, highlights the importance of the KDDCup'99 dataset, and offers a detailed analysis of classification techniques employed in intrusion detection. [2] As Internet connectivity grows and network traffic increases, the need for effective network intrusion detection systems (NIDS) becomes more critical to combating sophisticated attacks. These systems often rely on techniques like pattern recognition and data mining to classify network events as normal or malicious. The Hidden Naive Bayes (HNB) model is particularly useful for intrusion detection tasks with highly correlated features and large data stream volumes. Compared to traditional naive Bayes models and other advanced methods like support vector machines, the HNB model demonstrates superior performance in terms of accuracy, error rate, and misclassification cost. Specifically, it shows enhanced accuracy in detecting denial-of-service (DoS) attacks, highlighting its effectiveness in addressing complex intrusion scenarios. [3] Distributed denial of service (DDoS) attacks remains a persistent challenge for users and organizations. Security engineers strive to ensure service availability by using intrusion-detection systems (IDS) to detect and classify any unusual activity. These systems need to stay updated with the latest attack prevention methods to maintain service confidentiality, integrity, and availability. This paper addresses the need for updated datasets containing modern DDoS attacks across different network layers, such as SIDDoS and HTTP Flood. The study employs three well-known classification techniques: multilayer perceptron (MLP), Naïve Bayes, and random forest, and finds that MLP achieved the highest accuracy rate at 98.63%. [4] The workflow of the proposed method is setup as shown in the Figure 1, starting with data collection (KDD-99 Dataset), Pre-Processing: Training and testing dataset, building model and result analysis

3.1 KDD cup 1999 dataset Collection

In 1998, the DARPA Intrusion Detection Assessment Program was prepared and managed by MIT Lincoln Labs. Its purpose was to study and evaluate intrusion detection research. Standard data sets include various simulation intrusions in military network environments. The connection to the dataset includes a sequence of TCP packets beginning and ending at a well-defined time between the source IP address and the destination IP address using a welldefined protocol. Each connection is categorized as a normal or specific type of attack. Data sets are categorized into five sub-sets: denial-of-service attacks, local or remote network attacks, user / root attacks, sample attacks, and generic data. Each record is classified as normal or attack with exactly one type of attack. They are categorized as follows:

- Denial of service (DoS) Denial of Service (DOS) allows a legitimate user to gain access to the machine by creating too much or too much computer resources or memory for an attacker to handle legitimate requests.
- R2L (Local Remote Attack (User)) Local Remote Attack (R2L) is a type of attack in which an attacker can send packets to a computer over the network and then exploit a vulnerability in the computer to illegally attack local access. On the machine.
- Root User Attack (U2R) Root User Attack (U2R) is the attack class that an attacker first accesses a regular user account on a system. The vulnerability could be exploited to gain root access to the system.
- Monitoring (monitoring and other discovery) detection is an attack type in which an attacker scans the network for known information or vulnerabilities. An attacker with a map of systems and services available on the network will use the information to detect attacks.

3.2 Data Pre-processing:

Preprocessing involves Handling Null Values: - is null() method to check whether a null values is present in dataset . Label Encoder: le=label Encoder () method label_ encoder is used to transferring Categorical data into Numerical data

3.3 Feature Selection

Information Gain Ratio based feature selection: Features selected based on only information gain is biased towards attributes having many values. Information Gain Ratio (IGR) based Feature Selection removes this drawback by taking the splitting information of an attribute into account. Splitting information of an attribute is the entropy of pattern distribution into branches. Gain ratio of attribute decreases as value of split information increases.

Algorithm: 1. Start with the full set of

attributes (set containing all attributes of the dataset) and null selected feature set. 2. Calculate information gain ratio of each attribute. 3. Choose an attribute from the total set with the highest information gain ratio. 4. Split the dataset into sub datasets depending on the attribute values. 5. Add the attribute to selected feature set and remove from set of attributes. 6. Repeat step 2 to 5 for each of the sub-datasets with the set of attributes, if instance in a sub-dataset belongs to more than one class. 7. Output the selected feature set

Figure 2: Feature Selection

3.4 Model building The supervised machine learning algorithms [1] are those algorithms which needs external assistance. The input dataset is divided into train and test dataset. The train dataset has output variable which needs to be predicted or classified. All algorithms learn some kind of patterns from the training dataset and apply them to the test dataset for prediction or classification. The proposed method distinguish the normal traffic from malicious traffic using supervised algorithms such as C4.5 Decision tree and Naïve Bayesian Classifier

a. C4.5 Decision tree C4.5 Decision tree is one of the simple technique used in the machine learning and data mining. It is used as a predictive model in which observations about an item are mapped to conclusions about the item's target value. In the process of decision analysis, a decision tree can be used to represent decisions and decision making visually and explicitly. In this algorithm, the data set is learnt and modelled. Therefore, whenever a new data item is given for classification, it will be classified accordingly learned from the previous dataset

The steps of the algorithm are as follows:

1. If all the given training examples belong to the same class, then a leaf node is created for the decision tree by choosing that class.
2. For every feature 'a', calculate the gain ratio by dividing the information gain of an attribute with splitting value of that attribute. The formula for gain ratio is $\text{Gain Ratio } a(a) = \text{IG}(a) / \text{Split}(a)$ where, S is the set of all the examples in the given training set.

RESULT In this paper author is evaluating performance of two supervised machine learning algorithms such as SVM (Support Vector Machine) and ANN (Artificial Neural Networks). Machine learning algorithms will be used to detect whether request data contains normal or attack (anomaly) signatures. In above lines we can see normal is having id 0 and Anomaly has id 1 and goes on for all attacks. Before running code execute below two commands

Screen shots Double click on 'run.bat' file to get below screen

RESULT:

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Screen shots Double click on 'run.bat' file to get below screen



Figure1 1:

In above screen click on 'Upload NSL KDD Dataset' button and upload dataset



Figure2

In above screen I am uploading 'intrusion_dataset.txt' file, after uploading dataset will get below screen



Figure 3

Now click on 'Pre-process Dataset' button to clean dataset to remove string values from dataset and to convert attack names to numeric values

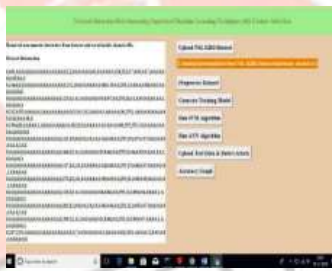


Figure4



From above graph we can see ANN got better accuracy compare to SVM, in above graph x-axis contains algorithm name and y-axis represents accuracy of that algorithms After pre-processing all string values removed and convert string attack names to numeric values such as normal signature contains id 0 and anomaly attack contains signature id 1. Now click on 'Generate Training Model' to split train and test data to generate model for prediction using SVM and ANN

Conclusion

In this paper Intrusion detection is considered as a classification problem where each record can be classified into normal or a particular kind of intrusion. Intrusion detection using machine learning has attracted more and more interests in recent years. As an important application of machine learning, an accurate intrusion detection model is built by choosing an effective classification approach. This paper shows the comparison of the most well-known classification algorithms like C4.5 decision trees and Naive Bayes has been carried out. These algorithms are tested with the KDD data-set. Effective classifier is identified by comparing the performances based on the accuracy and confusion matrix. Performance calculation is done by considering only the important attributes for the intrusion detection. From the observed results it can be concluded that the C4.5 decision trees classifier outperforms other classifiers for the considered data-set and parameters. It has the accuracy of 99%.

References

- [1] Ahmed CA, C A, Mohammad M, Mohammed A, "An Effective Mechanism to Mitigate Packet Dropping Attack from MANETs using Chaotic Map based Authentication Technique", Recent Patents on Engineering, vol 18, no 3, pp. 66-76, 2024.
- [2] Mohammed A, Mohammad M, "How AI Algorithms Are Being Used in Applications". In International Conference on Soft Computing and Signal Processing, pp. 41-53, 2022 Jun 24.
- [3] Ashley Chonka, Jaipal Singh, Wanlei Zhou, "Chaos theory-based detection against network mimicking DDoS attacks," IEEE Communications Letters, vol 13, no 9, 2009.

- [4] Mohammed, A., Singh RP, Ahmad SJ, “Enhance Multiple Moving Target Detection in Doppler-Tolerant Radar Using IRAESC Technique”, In Innovations in Electronics and Communication Engineering: Proceedings of the 6th ICIECE, pp. 417-426, 2018 Aug 29.
- [5] Mohammad M, Mohammed A, “Smart cities implementation: australian bushfire disaster detection system using IoT innovation”, Smart Innov, Syst Technol 313 [Internet], 2023
- [6] Atheeq C, Ali L, Altaf C, Mohammed A “Mitigating man-in-the-middle attack in UAV network using authentication mechanism based on chaotic maps” In Emerging Trends in Computer Science and Its Application pp. 117-123, 2025 Apr 8.
- [7] Mohammad M, Mohammed A, KBAR G, YACOUB A, “The role of ICT in educational settings leadership in learning and teaching with ICT: The case of Qatar”.
- [8] Y. Xiang ; Y. Lin , W.L. Lei , S.J. Huang “Detecting DDOS attack based on network self similarity,” IEE Proceedings – Communications, vol 151, no. 3, 2014.
- [9] Mohammad M, Mohammed A, Atheeq C, “Prior bush fire identification mechanism based on machine learning algorithms full text”, 2022
- [10] Abdullahi MH, Mohammed A, “Influence Of Cinnamon And Garlic On Chemical And Microbial Properties Of Beef Minced Meat”, Journal of Agricultural Economics, Environment and Social Sciences vol 11, no 2, pp.27-33, . 2025.
- [11] Mohammed A, Abdullahi M, Muhammad D, Ibrahim I, Mohammed IA, Idris IB. “Assessment of Internet Security Skills Need of Computer Education Students for Effective Communication and Networking in Universities in Gombe State, Nigeria”, Kashere Journal of Education, vol 7, no 2, pp. 112-121, 2024.
- [12] Banitalebi Dehkordi, Mohammad Afsaneh, Reza Soltanaghaei and Farsad Zamani Boroujeni, "The DDoS attacks detection through machine learning and statistical methods in SDN", The Journal of Supercomputing, vol. 77, no. 3, pp. 2383-2415, 2021.
- [13] Mohammad, M., Mohammed, A., & Vulla, H., Atheeq, C., “Quantum-Adaptive Neural Defense: A Self-Learning Ddos Mitigation Framework Using Entropy-Based Traffic Fingerprinting And Real-Time Behavioral Analysis”, Power System Protection and Control, vol 53, no 4, pp. 319-334, 2025.
- [14] Aqeel Sahi , David Lai , Yan Li , Mohammed Diykh, “An Efficient DDoS TCP Flood Attack Detection and Prevention System in a Cloud Environment,” IEEE Access , Volume: 5, April 2017.
- [15] Mohammed, A., Mohammad M., Butt, Atif & Atheeq, C. “HIVE-SIGHT: A Swarm-Intelligence Framework for Anticipating and Mitigating Distributed Denial-of-Service Attacks at Global Scale”, Acta Scientiae, vol 26. 2025.
- [16] Mohammed A., Mohammad M, & Atheeq, C., “Ecosystems: A Comprehensive Review of Transformer Architectures, Quantum Machine Learning, and Federated Explainable AI”, Acta Scientiae, 2025.
- [17] Atheeq, C., Mohammad, M., Mohammed, A., & Mohammad, A, “A study on mitigation of attacks from MANETS based on authentication mechanism”, 2023.
- [18] Mohammad, M., & Mohammed, A., “Pre-and post-assessment of COVID-19 in academic learning: A state of the art assessment”, 2022.
- [19] Wang Bing, Zheng Yao, Lou Wenjing, “DDoS attack protection in the era of cloud computing and software defined networking”. Computer Networks, 2015, 308—319.